R basics

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Basic

Directory

To know the working directory

```
getwd()
```

```
[1] "G:/Lesson/Done/Research/R Language/R-Language/Basic"
```

Set the working directory

```
#setwd("G\\....\\...\like-this.csv")
```

Show the file in the working directory

```
list.files()
```

```
[1] "R-basics.rmarkdown" "R basics.qmd"       "R basics.rmarkdown"
```

Conditional Sentance

```
a <- 10
b <- 12
if(a>b)
{print("a is greater than b")} else if(a==b)
{print("a and b are equal")} else
{print("b is greater than a")}
```

```
[1] "b is greater than a"
```

Loop

```
for(x in 1:10)
{print(x)}
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
[1] 6
[1] 7
[1] 8
[1] 9
[1] 10
```

Function

```r
avarage <- function(x,y,z){
ava=(x+y+z)/2
print(ava)
}

avarage(2,3,4)
```

```
[1] 4.5
```

Data Type

```r
a <- 10
typeof(a)
```

```
[1] "double"
```

```r
b <- "swarup"
typeof(b)
```

```
[1] "character"
```

```r
c <- TRUE
typeof(c)
```

```
[1] "logical"
```

```r
d <- 10L
typeof(d)
```

```
[1] "integer"
```

```r
e <- 5i
typeof(e)
```

```
[1] "complex"
```

Sub Setting

```r
number <- 1:100

odd_number <-  seq(1,99,2)
from100_1 <-  seq(100,1,-1)

from100_1
```

```
  [1] 100  99  98  97  96  95  94  93  92  91  90  89  88  87  86  85  84  83
 [19]  82  81  80  79  78  77  76  75  74  73  72  71  70  69  68  67  66  65
 [37]  64  63  62  61  60  59  58  57  56  55  54  53  52  51  50  49  48  47
 [55]  46  45  44  43  42  41  40  39  38  37  36  35  34  33  32  31  30  29
 [73]  28  27  26  25  24  23  22  21  20  19  18  17  16  15  14  13  12  11
 [91]  10   9   8   7   6   5   4   3   2   1
```

```r
odd_number
```

```
 [1]  1  3  5  7  9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49
[26] 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97 99
```

```r
odd_number[10:20]
```

```
 [1] 19 21 23 25 27 29 31 33 35 37 39
```

```r
odd_number[c(12,32,50,99)]
```

```
[1] 23 63 99 NA
```

```r
odd_number>20
```

```
 [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE
[13]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
[25]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
[37]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
[49]  TRUE  TRUE
```

Variable type change

```r
double <- 3.1416
typeof(double)
```

```
[1] "double"
```

```r
int <- as.integer(double)
int
```

```
[1] 3
```

```r
typeof(int)
```

```
[1] "integer"
```

```r
srt <- as.character(double)
srt
```

```
[1] "3.1416"
```

```r
typeof(srt)
```

```
[1] "character"
```

```r
logi <- as.logical(double)
logi
```

```
[1] TRUE
```

```r
typeof(logi)
```

```
[1] "logical"
```

Creating Dataset in R

Simple

```
student_data <- data.frame (name=c("shadhin","joy","durjoy"),roll=c(23,45,32))
```

Creating with rep function

hare 'rep' means repeated This I will use in the One Way ANOVA

```
data_rep <- data.frame(method=rep(c("No 1","No 2","No 3"),each=4), marks=c(1,2,3,4,5,6,7,8,9
data_rep
```

```
   method marks
1    No 1     1
2    No 1     2
3    No 1     3
4    No 1     4
5    No 2     5
6    No 2     6
7    No 2     7
8    No 2     8
9    No 3     9
10   No 3     1
11   No 3     2
12   No 3     3
```

With metrix

```
met <- matrix(c(23,43,24,24,23,34,32,35,32,35,35,35,34,NA,36), nrow=5,ncol=3,byrow=TRUE)
met
```

```
     [,1] [,2] [,3]
[1,]   23   43   24
[2,]   24   23   34
[3,]   32   35   32
[4,]   35   35   35
[5,]   34   NA   36
```

```
is.data.frame(met)
```

```
[1] FALSE
```

```r
length(met)
```

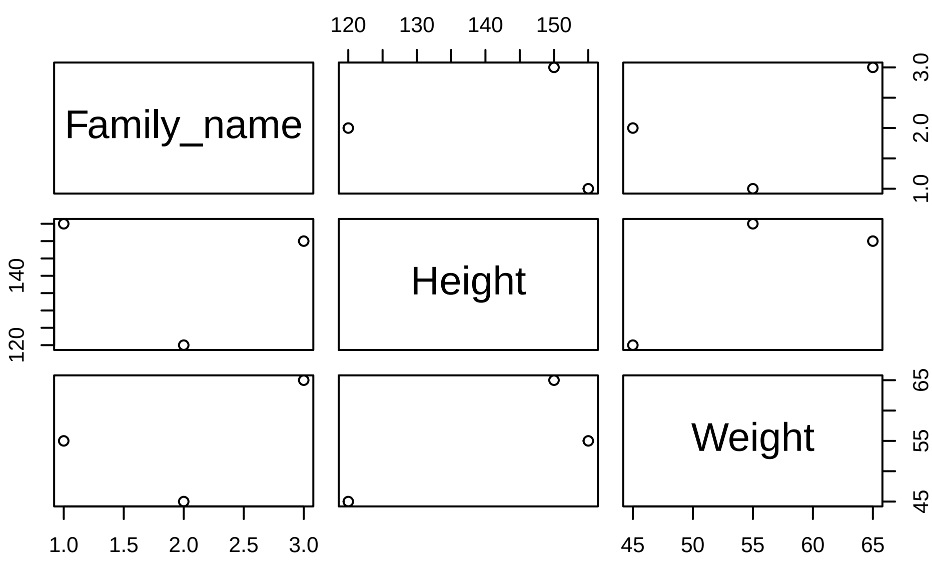
```
[1] 15
```

Practice 1 - BMI Calculation

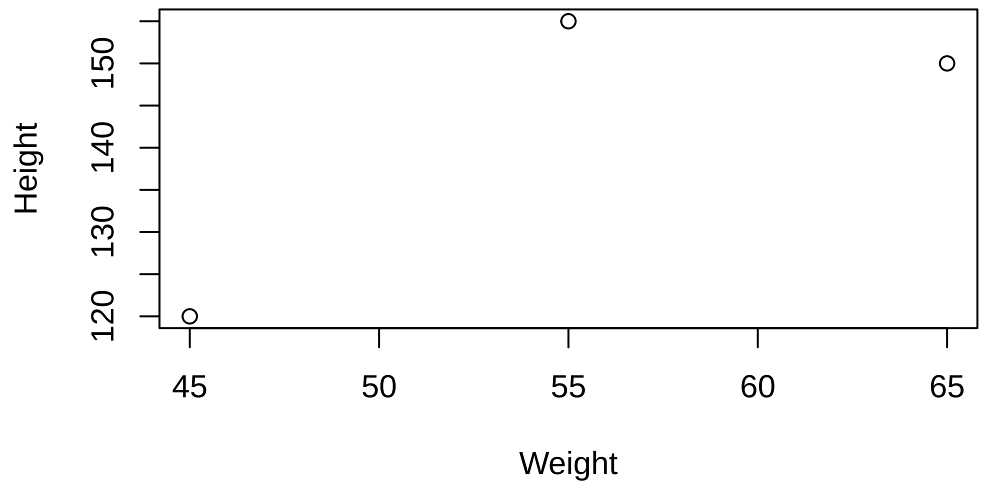
```r
#column info
Family_name <-c("Durjoy", "Swapon", "Lipy")
Height <- c(155,150,120) #in cm
Weight <- c(55,65,45) #in kg

#Making it a data set or data frame
Family_info <- data.frame(Family_name, Height, Weight)

#Some visualization
plot(Family_info)
```




```r
plot(Height~Weight, Family_info)
```




```
#data filter***
Family_info[Family_info$Height==155,]
```

```
  Family_name Height Weight
1      Durjoy    155     55
```

```
# Creating a new column of BMI
Family_info$BMI <-Family_info$Weight/(Family_info$Height*0.01)^2
Family_info
```

```
  Family_name Height Weight      BMI
1      Durjoy    155     55 22.89282
2      Swapon    150     65 28.88889
3        Lipy    120     45 31.25000
```

```
# Adding a new category***
Family_info$Health_status <- ifelse(Family_info$BMI<=18.5,"Underweight",ifelse(Family_info$BMI
Family_info
```

```
  Family_name Height Weight      BMI Health_status
1      Durjoy    155     55 22.89282 NORMAL WEIGHT
```

```
2       Swapon    150      65 28.88889    OVERWEIGHT
3         Lipy    120      45 31.25000       OBESITY
```